



UNIVERSITY OF  
BIRMINGHAM

# Superconductivity

## Lecture 22 – “Explained II”

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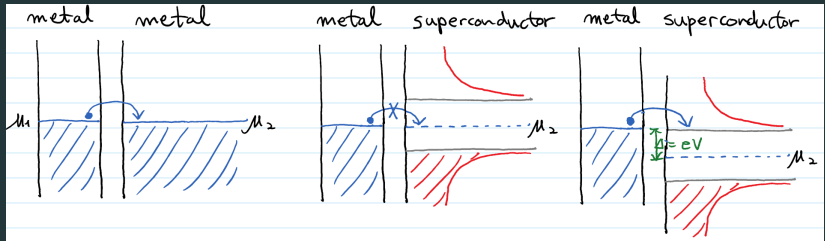
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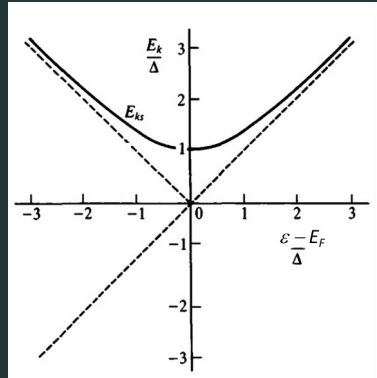
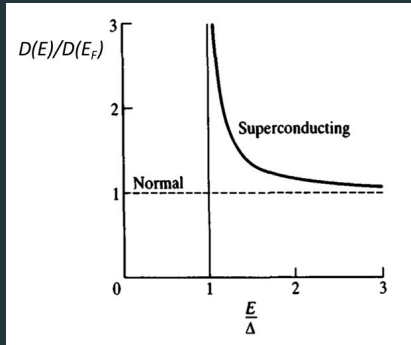
School of Physics & Astronomy

# Previously, tunneling

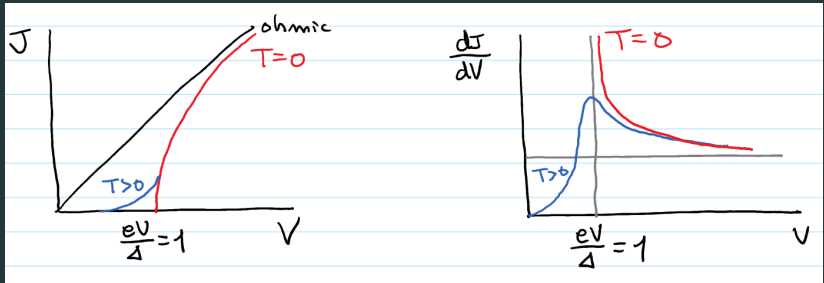


**Figure 1:** Energy versus density of states for three different contacts of metal-metal, metal-superconductor and superconductor at zero bias, and metal-superconductor at threshold bias.

## Previously, density of states



## Previously, normal metal to superconductor

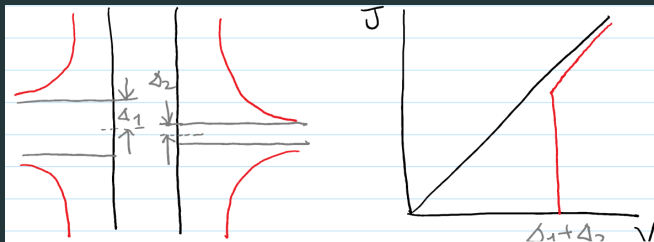


**Figure 2:** Current-voltage characteristic (left) and differential conductance versus voltage (right).

Conductance ( $T = 0$ )

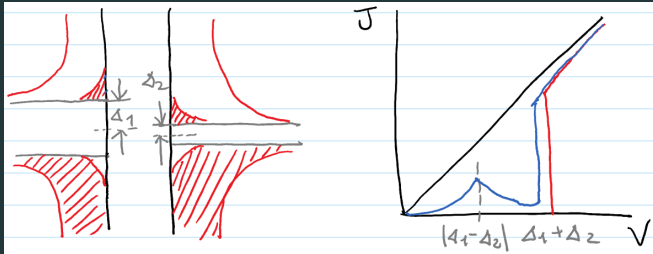
$$G_{ns} = \frac{dI}{dV} = G_{nn} \frac{D_s(eV)}{D_n(E_F)} \quad (1)$$

## Two superconductors with different gaps ( $T = 0$ )



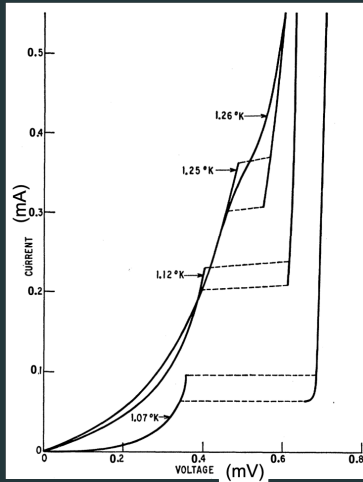
**Figure 3:** Density of states (left) and I-V characteristic of two different superconductors at zero temperature.

## Two superconductors with different gaps ( $T \neq 0$ )



**Figure 4:** Density of states (left) and  $I$ - $V$  characteristic of two different superconductors at finite temperature.

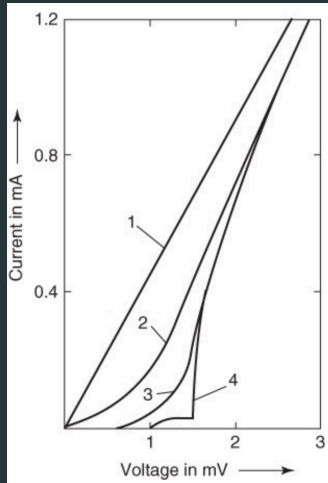
# Exercise 1: Tunneling experiment S-I-S



**Figure 5:** *I-V* tunneling characteristic of a S-I-S junction. (Neglect the hysteric behaviour marked by dashed lines.)

1. Explain why the curve changes its shape as temperature changes in the way as observed.
2. Estimate the values of the most relevant physical quantity of the two superconductors out of which the junction is made.

## Exercise 2: Tunneling experiment S-I-S

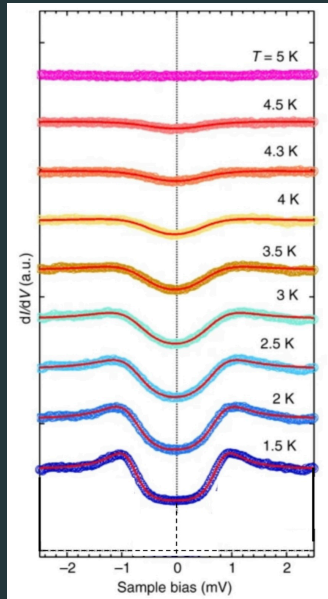


**Figure 6:** Voltage-current characteristic of an Al-Al<sub>2</sub>O<sub>3</sub>-Pb tunnel junction measured at different temperatures.

Curve 1 measured at 10 K, curve 2 at 4.2 K, curve 3 at 1.6 K and curve 4 at 1.0 K. Noting that Pb has a higher  $T_c$  than Al, use the figure to

1. Estimate  $\Delta$  at 1.0 K and  $T_c$  for both Al and Pb,
2. Briefly explain how it has been estimated.
3. Deduce the relationship between  $\Delta$  at zero temperature and  $T_c$  and briefly justify your answer.

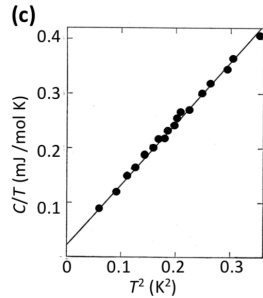
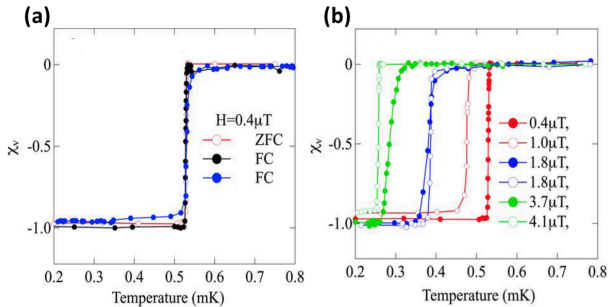
## Exercise 3: Differential conductance



1. Estimate the superconducting gap parameter,  $\Delta$ , at zero temperature (briefly discuss your method of estimation).
2. Use the available data to discuss whether the result agrees with the BCS prediction.

Note that the data are shifted vertically for better visibility.

# Exercise 4: Heat capacity



1. Estimate the Debye temperature  $\Theta_D$ , noting that the lattice specific heat contribution is given by  $(12/5)\pi^2 R(T/\Theta_D)^3$ .
2. Using experimental data available from the Figures and relevant formula within the BCS framework to discuss whether this superconductor is in the weak or strong coupling limit.

## Solution to Exercise 1 (1/2)

The cusp moves to a lower voltage as temperature decreases, and this implies that the gap of one superconductor increases as temperature is lowered while the gap of the other remains rather temperature independent.

Thus in the former superconductor, the measured temperature is not much below  $T_c$ . In the latter superconductor, on the other hand, the measured temperature is likely already very low compared to its  $T_c$

(For your interest, the junction was Al-alumina-In, where  $T_c$  of bulk Al and In are 1.2 K and 3.4 K, respectively.)

## Solution to Exercise 1 (2/2)

At 1.25 K,  $\Delta_1 - \Delta_2 = 0.47$  mV and  $\Delta_1 + \Delta_2 = 0.5$  mV. So  $\Delta_1 = 0.485$  mV and  $\Delta_2 = 0.015$  mV.

At 1.12 K,  $\Delta_1 - \Delta_2 = 0.40$  mV and  $\Delta_1 + \Delta_2 = 0.6$  mV. So  $\Delta_1 = 0.5$  mV and  $\Delta_2 = 0.1$  mV

At 1.07 K,  $\Delta_1 - \Delta_2 = 0.35$  mV and  $\Delta_1 + \Delta_2 = 0.68$  mV. So  $\Delta_1 = 0.515$  mV and  $\Delta_2 = 0.165$  mV

## Solution to Exercise 2 (1/2)

At 10 K,  $V$  is linear to  $I$ , so the temperature is higher than  $T_c$  of both superconductors.

At 1.0 K,  $|\Delta_1 - \Delta_2| = 1.25$  meV and  $|\Delta_1 + \Delta_2| = 1.5$  meV.  $\Delta_1 = 1.375$  and  $\Delta_2 = 0.125$  meV.

At 4.2 K,  $V$  vs  $I$  shows deviation from linearity, implying  $T_c$  of one of the superconductors is higher than 4.2 K. This allows us to estimate that  $T_c = 7(3)$  K for one of the superconductors.

It is only at 1.0 K when the cusp is visible while not clear already at 1.6 K. This allows us to estimate the  $T_c = 1.3(3)$  K for the other superconductor.

(For your interest,  $T_c = 7.2$  K for Pb and 1.2 K for Al. Compare with our estimation in this exercise,  $T_c = 7(3)$  and 1.3(3)K, respectively)

## Solution to Exercise 2 (2/2)

To establish a relationship between  $\Delta$  and  $T_c$ , one can only use one of the superconductors which has a significantly higher  $T_c$  compared to  $\Delta$  at 1.0 K that may be approximated as  $\Delta$  at absolute zero.

Since  $k_B T_c = 8.6 \times 10^{-5} \times 7 = 0.6 \text{ meV}$ , and  $2\Delta = 2.7 \text{ meV}$ , one finds that  $2\Delta = 4.5 k_B T_c$ .

(For your interest, the BCS theory tells us  $2\Delta(0) = 3.52 k_B T_c$ )

## Solution to Exercise 3

It would be better to use the data at the lowest measured temperature of 1.5 K. The peak position implies that the gap size is about 1 meV. If one takes into account thermal smearing, the actual  $\Delta_0$  would be around 0.8 meV. From the figure, one can expect that the transition temperature is about 4.7 K. Then one can show that  $\Delta_0 \simeq 1.9k_B T_c$ , in good agreement with the BCS prediction of  $\Delta_0 \simeq 1.76k_B T_c$

## Solution to Exercise 4

From the figure, one can deduce  $C = 0.02T + 1.1T^3$ , and then calculate that  $\Theta_D = 120$  K. From the susceptibility data, one obtains  $T_c = 0.53$  mK. Then, substituting to the BCS expression for  $T_c$  and  $\Theta_D$ , one can evaluate that  $DV \ll 1$ , which support that the observed superconductivity is in the weak coupling limit.